

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Piping Class Specification
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Piping Class Specification

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Revision History

Rev No.	Date	Description of Revision
R01	01-06-22	Issued for Client's Review
R02		
A01		

Review Code & Status (to be filled by Client)

☐ Code 1	Approved – Further work may proceed.	
☐ Code 2	Approved with comments. – Incorporate Comments. Up-Rev & Resubmit. Work May Proceed	
☐ Code 3	Reviewed – Incorporate Comments. Up-Rev & Resubmit. Work May Not Proceed	
☐ Code 4	Retained for information only	
☐ Code 5	Review Not Required – Do not Resubmit	
Date:	Position:	Sign:

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-015	Specification for Piping
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005	Piping Classes
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-010	Specification for Pipes & Fittings
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-013	Specification for Flanges
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-012	Specification for Bolts and Nuts
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-014	Specification for Gaskets
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003	Specification for Piping Valves
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-011	Specification for Pipe Support.
EA-BFPCL-GPMC-GGPL-FEED-MPP-RPT-005	Piping wall thickness calculation

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Abbreviations

Abbreviation	Meaning
AIV	Acoustic Induced Vibration
API	American Petroleum Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
BB	Bolted Bonnet
BC	Bolted Cover
BE	Bevelled End
BW	Butt Welding
CL	Class
CNAF	Compressed Non-Asbestos Fiber
CONC	Concentric (reducer)
Cr	Chromium
CRA	Corrosion Resistant Alloy
CS	Carbon Steel
DBLE	Double
ECC	Eccentric (reducer)
EEMUA	The Engineering Equipment and Materials users Association
EFW	Electric Fusion Welded
ENP	Electroless Nickel Plated
EXTD	Extended (Bonnet)
FF	Flat Face
FIV	Flow Induced Vibration
FLGD	Flanged
FLTING	Floating
GRE	Glass Reinforced Epoxy
HB	Hardness Brinell
HF	Hard Facing of Cobalt-Chromium Alloy

HNDWL	Handwheel
ISO	International Organization for Standardization
LR	Long Radius
LTCS	Low Temperature Carbon Steel
MESC	Material & Equipment Standard and Code
MAN STD	Manufacturer Standard
Mo	Molybdenum
MSS	Manufacturers Standardization Society
MTD	(Trunnion) Mounted
Mmscf/d	Million standard cubic feet per day
NACE	National Association of Corrosion Engineers
Ni	Nickel
NPS	Nominal Pipe Size
NPT	National Pipe Taper
NPTF	Nominal Pipe Taper Female
OPR	Operated
OS&Y	Outside Screw spindle and Yoke
PE	Plain End
PTFE	Polytetrafluoroethylene (Teflon)
PVDF	Polyvinylidene Fluoride
RF	Raised Face
RJ	Machined face for ring Joint
RSNRO	Rising Stem Non Rising Operator
RSRO	Rising Stem Rising Operator
SAW	Submerged Arc Welded
SCRN	Screen
SP	Standard Practice (MSS)
SPB	Split Body
SS	Stainless Steel

STD	Standard
STLS STL	Stainless Steel
SW	Socket Welding
TCC	Tungsten Carbide Coating
TE	Threaded End
THK	Thickness
THRD	Threaded
TPE	Top Entry
TPLE	Triple
UNS	Unified Numbering System
W/	Complete With
WB	Welded Bonnet
WC	Welded Cover
WN	Welding Neck

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

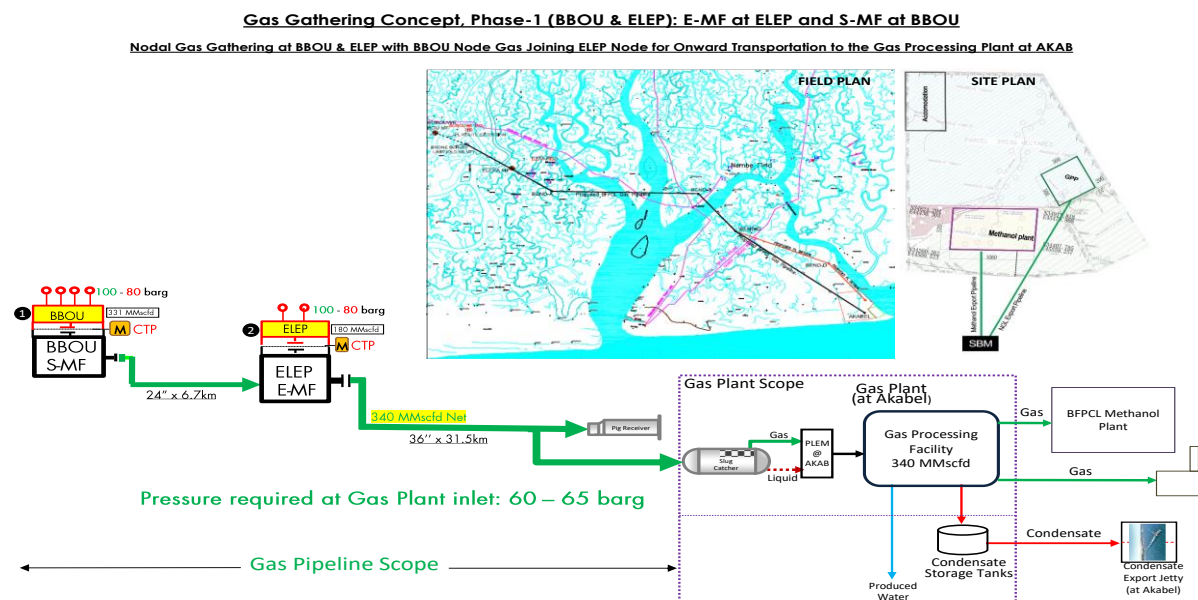


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification presents the classification and the minimum technical requirements for pipes, fittings, flanges, valves, and other piping components used for process and utility piping systems designed to ASME B31.3 for BFPCL Gas gathering Pipeline project phase 1.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 31.38.01.25-Gen.	Piping – Process Design Requirements
Dep 31.38.01.10-Gen	Piping Classes – Basis of Design
DEP 31.38.01.24-Gen.	Piping – Engineering and Layout Requirements
DEP 31.38.01.31-Gen	Shop And Field Fabrication of Piping
DEP 31.38.01.84-Gen	Piping Classes
MESC SPE 76/210	Carbon Steel Forgings (Amendments/Supplements to ASTM A 105)
MESC SPE 76/101	Flanges (Amendments/Supplements to ASME B16.47)
MESC SPE 76/212	Alloy And Stainless-Steel Forgings (Amendments/Supplements to ASTM A 182)
MESC SPE 76/100	Flanges (Amendments/Supplements to ASME B16.5)
API SPEC 5L	Specification for Line Pipe
API SPEC 6A	Specification for Wellhead and Christmas Tree Equipment
API STD 594	Check Valves: Flanged, Lug, Wafer, and Butt-welding
API STD 599	Metal Plug Valves - Flanged, threaded and welding ends
API STD 600	Steel Gate Valves - Flanged and Butt-welding Ends, Bolted Bonnets
API STD 602	Gate, Globe, and Check Valves for Sizes DN 100 (NPS 4) and Smaller for the Petroleum and Natural Gas Industries
API STD 609	Butterfly Valves: Double-flanged, Lug- and Wafer-type
API STD 623	Steel Globe Valves - Flanged and Butt-welding Ends, Bolted Bonnets
ASME B16.10	Face to Face and End to End Dimensions of Valves
ASME B16.11	Forged Fittings, Socket-Welding and Threaded
ASME B16.20	Metallic Gaskets for Pipe Flanges - Ring-Joint, Spiral-Wound, and Jacketed
ASME B16.21	Non-metallic Flat Gaskets for Pipe Flanges
ASME B16.25	Butt-welding Ends

ASME B16.34	Valves-Flanged, Threaded, and Welding End
ASME B16.36	Orifice Flanges
ASME B16.47	Large Diameter Steel Flanges: NPS 26 through NPS 60 Metric/Inch Standard
ASME B16.48	Line Blanks
ASME B16.5	Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch Standard
ASME B16.9	Factory Made Wrought Buttwelding Fittings
ASME B18.2.2	Nuts for General Applications: Machine Screw Nuts, Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)
ASME B31.3	Process Piping
ASME B31.8	Gas Transmission and Distribution Piping Systems
ASME B36.10M	Welded and Seamless Wrought Steel Pipe
ASME B36.19M	Stainless Steel Pipe
ASME BPVC Section II	Materials
ASME BPVC Section VIII Div 1	Rules for Construction of Pressure Vessels - Division 1
ASME BPVC Section VIII Div 2	Rules for Construction of Pressure Vessels - Division 2: Alternative Rules
ASTM A105	Standard Specification for Carbon Steel Forgings for Piping Applications
ASTM A106	Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service
ASTM A182	Standard Specification for Forged or Rolled Alloy and Stainless-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service
ASTM A193	Standard Specification for Alloy-Steel and Stainless-Steel Bolting for High Temperature or High-Pressure Service and Other Special Purpose Applications
ASTM A194	Standard Specification for Carbon Steel, Alloy Steel, and Stainless-Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both
ASTM A234	Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service
ASTM A312	Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless-Steel Pipes
ASTM A320	Standard Specification for Alloy-Steel and Stainless-Steel Bolting for Low-Temperature Service

ASTM A333	Standard Specification for Seamless and Welded Steel Pipe for Low-Temperature Service and Other Applications with Required Notch Toughness
ASTM A350	Standard Specification for Carbon and Low-Alloy Steel Forgings, Requiring Notch Toughness Testing for Piping Components
ASTM A351	Standard Specification for Castings, Austenitic, for Pressure-Containing Parts
ASTM A352	Standard Specification for Steel Castings, Ferritic and Martensitic, for Pressure-Containing Parts, Suitable for Low-Temperature Service
MSS SP-83	Class 3000 and 6000 Pipe Unions, Socket Welding and Threaded (Carbon Steel, Alloy Steel, Stainless Steels, and Nickel Alloys)
MSS SP-95	Swage(d) Nipples and Bull Plugs
MSS SP-97	Integrally Reinforced Forged Branch Outlet Fittings - Socket Welding, Threaded, and Buttwelding Ends

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. General considerations

3.1. General

- Design life of the piping system in operation is 30 years.
- Mechanical design limits for material classes are determined as per ASME B16.5

3.2. Material notes on Valves

- For detailed information on valve design, refer to the document EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003 Specification for Piping Valves and valve data sheets.

3.3. Material notes for pipe

- Pipes 2" or more in diameter shall be delivered with ends bevelled as per ASTM B16.25. Pipes less than 2" in diameter shall have plain ends square cut, with burrs removed.
- The following pipe sizes shall not be used: NPS 2-1/2, 5, 7, 22
- Pipes dimension is in accordance with ASME B36.10M
- Minimum wall thickness for NPS 2" and below shall be SCH 80.
- Refer to EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-010 for Specification for Pipes & Fittings

3.4. Material notes for flanges

- Class 150# flanges shall be Raised face. While class 900# and 1500# flanges shall be RJ in accordance with ASME B16.5.
- Refer to EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-013 for specification for Flanges

3.5. Material notes for fittings

- Fittings shall be seamless for all sizes less than NPS 20"
- Plugs shall be forged, solid, and long round head.
- Butt-weld end preparations for welding shall be in accordance with ANSI B16.25.

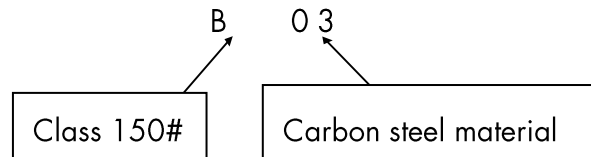
3.6. Material notes for Bolting, Gaskets, and Spectacle Blind

- All stud bolts shall be threaded full length, complete with two heavy hexagon nuts.
- Bolting to be UNC threads (up to 1") and 8UN (1-1/8" and larger) with UN heavy hexagon nuts. Dimensions bolts to B16.5, nuts to ANSI B18.2.2. Thread to ANSI B1.1, bolts Class 2A, nuts class 2B. Nuts to be chamfered and faced.
- All bolts and studs must be supplied already assembled with relevant number of nuts.
- All spiral wound gaskets shall be marked, and colour coded in accordance with ASME B16.20. Colours shall be as per Table 19 with colour coding as per paragraph 3.4.3. Marking shall be as per paragraph 3.4.1 of ASME B16.20.

4. Class numbering principles

The class number shall consist of a capital letter representing the ASME or API flanges rating, and a two-digit number representing the main composition of the material used for the pipes, fittings, flanges, valves, etc., of the piping material class in question.

Example:



Letters representing the ASME and API SPEC 6A flanges rating of the classes			
Rating	150#	900#	1500#
Letter	B	C	D

5. LIST OF PIPING CLASS

CLASS	FLANGE RATING	FLANGE FACE	MATERIAL (corrosion in mm)	SYMBOL	SERVICES	MAX. DESIGN CONDITIONS		NPS max (inches)
						PRESSURE (barg)	TEMPERATURE (°C)	
B 03	ASME CLASS 150	Raised Face (RF)	CARBON STEEL (3.0 mm)	HG HL VG CD OD	Hydrocarbon Gas Hydrocarbon Liquid Low Vent Gas Close Drain Open Drain	19.6 13.8	-29 +200	42
B 06	ASME CLASS 150	Raised Face (RF)	CARBON STEEL (6.0 mm)	HG HL VG CD OD	Hydrocarbon Gas Hydrocarbon Liquid Low Pressure Flare Close Drain Open Drain	19.6 13.8	-29 +200	42
B 13	ASME CLASS 150	Raised Face (RF)	CARBON STEEL, GALVANISED (3 mm)	UW	Utility Water	19.6 13.8	-29 +200	2
B 23	ASME CLASS 150	Raised Face (RF)	LOW TEMPERATURE CARBON STEEL (3 mm)	HG HL VG CD OD	Hydrocarbon Gas Hydrocarbon Liquid Low Vent Gas Close Drain Open Drain	19.6 13.8	-46 +200	42
B 26	ASME CLASS 150	Raised Face (RF)	LOW TEMPERATURE CARBON STEEL (6.0 mm) / <u>NACE</u>	HG HL VG CD OD	Hydrocarbon Gas Hydrocarbon Liquid Low Vent Gas Close Drain Open Drain	19.6 13.8	-46 +200	46
C 03	ASME CLASS 900	Octagonal Ring Joint (RJ)	CARBON STEEL (3.0 mm)	HG HL	Hydrocarbon Gas Hydrocarbon Liquid	153.2 131.4	-29 +200	
C 06	ASME CLASS 900	Octagonal Ring Joint (RJ)	CARBON STEEL (6.0 mm)	HG HL	Hydrocarbon Gas Hydrocarbon Liquid	153.2 131.4	-29 +200	46
C 23	ASME CLASS 900	Octagonal Ring Joint (RJ)	LOW TEMPERATURE CARBON STEEL (3 mm)	HG HL	Hydrocarbon Gas Hydrocarbon Liquid	153.2 131.4	-46 +200	46
C 25	ASME CLASS 900	Octagonal Ring Joint (RJ)	LOW TEMPERATURE CARBON STEEL (5.0 mm)	HG HL	Hydrocarbon Gas Hydrocarbon Liquid	153.2 131.4	-46 +200	46

C 50	ASME CLASS 900	Octagonal Ring Joint (RJ)	STAINLESS STEEL - UNS N08904 + TYPE 316L (0 mm)	CI DE AF	Corrosion inhibitor Demulsifier Anti-foam	124.1 87.5	-46 +200	2
C70	ASME CLASS 900	Octagonal Ring Joint (RJ)	DUPLEX STAINLESS STEEL - UNS S31803 (0 mm)	HG HL CD	Hydrocarbon Gas (H ₂ S + CO ₂ heavy corrosion) Hydrocarbon Liquid (H ₂ S + CO ₂ heavy corrosion) Closed Drain	155.1 128	-46 +200	46
D70	ASME CLASS 1500	Octagonal Ring Joint (RJ)	DUPLEX STAINLESS STEEL - UNS S31803 (0 mm)	HG HL CD	Hydrocarbon Gas (H ₂ S + CO ₂ heavy corrosion) Hydrocarbon Liquid (H ₂ S + CO ₂ heavy corrosion) Closed Drain	258.6 213.3	-46 +200	36

6. Piping materials class Table

PIPING MATERIAL CLASS																														brass fertilizer																																																																																																																		
FEED for Gas Gathering Pipelines & Associated Infrastructure(Phase 1)																																																																																																																																																
MATERIAL: CARBON STEEL															FLANGE RATING: ASME CLASS 150 RF										CLASS B 03																																																																																																																							
SERVICES: HYDROCARBON GAS HYDROCARBON LIQUID LOW PRESSURE FLARE															PRESSURE / TEMPERATURE LIMITS (MATL GRP 1.1)																																																																																																																																	
															PRESS. (BARG)					19.6										19.6					17.7					13.8																																																																																																								
															TEMP. (°C)					-29										+38					+100					+200																																																																																																								
															CORROSION										3.0 mm					SHEET: 1					OF 1					REVISION: 1																																																																																																								
NPS					0.5					0.75					1					1.5					2					3					4					6					8					10					12					14					16					18					20					24					26					28					30					32					34					36					38					40					42					44					46					NOTES				
SCHEDULE					160					160					160					160					XS					STD					STD					20					20					30					20					20					20					STD					STD					STD					40					XS					XS					XS					XS					XS					108.109																													
WALL THICK (mm)					4.78					5.56					6.35					7.14					5.49					6.02					7.11					12.70					12.70					13.50					14.80					16.90					18.10					20.10					24.20					26.20					28.20					30.20					32.20					34.20					36.2					38.2					40.3					42.3					12.7					12.7					12.7					108.109				
OD (mm)					21.3					26.7					33.4					48.3					60.3					88.9					114.3					168.3					219.1					273					323.9					355.6					406.4					457					508					610					660					711					762					813					864					914					965					1016					1067					1118					1168									
PIPE					SEAMLESS, ASME B36.10, API SPEC 5L GRADE B PSL 2															WELDED, SAW 100% RADIOGRAPHY, ASME B36.10, API SPEC 5L GRADE B PSL 2															110																																																																																																													
ENDS					PE															BE																																																																																																																												
FLANGES					FLANGE					SW, RF, CL150, ASME B16.5, A105N										WN, RF, CL150, ASME B16.5, HUB THK AS PER PIPE SCHEDULE, ASTM A105N										WN, RF, CL150, ASME B16.47 SERIES A, HUB THK AS PIPE SCHEDULE, ASTM A105N										250																																																																																																								
					BLIND					BLIND, RF, CL150, ASME B16.5, ASTM A105N										BLIND, RF, CL150, ASME B16.47 SERIES A, ASTM A105N										250																																																																																																																		
					ORIFICE					ORIFICE, WN, RF, CL300, ASME B16.36, ASTM A105N										250																																																																																																																												
FITTINGS					ELBOW					45/90 DEG SW, CL6000, ASME B16.11, A105N										45/90 DEG LR, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A234 WPB										45/90 DEG LR, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A234 WPBW										211.212																																																																																																								
					TEE					TEE, SW, CL6000, ASME B16.11, A105N										EQUAL TEE/REDUCING TEE, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A234 WPB										EQUAL TEE/REDUCING TEE, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A234 WPBW										202.211																																																																																																								
					REDUCER					RED CPLG, SW, CL6000, ASME B16.11, A105N										ECC/CONC, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A234 WPB										ECC/CONC, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A234 WPBW										211																																																																																																								
					CAP					CAP, SW, CL6000, ASME B16.11, A105N										PIPE CAP, SEAMLESS, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A234 WPB																																																																																																																												
					OLET					SOCKOLET, SW, CL6000, ASME B16.11, MSS SP-97, A105N										WELDOLET, MSS SP-97, ASTM A105N										202																																																																																																																		
					SWEEPOLET															SWEEPOLET, MANUFACTURER STANDARD DESIGN, ASTM A105N										202																																																																																																																		
VALVES					GATE					SW, CL800, API STD 602, A105N BODY, TRIM 16, OS&Y, BB, HNDWHL OPR										FLGD, RF, CL150, API STD 600, ASME B16.10, ASTM A216 WCB BODY, OS&Y, BB, FLEXIBLE WEDGE										507																																																																																																																		
										TRIM 16					TRIM 8																																																																																																																																	
										HNDWHL OPR										HNDWHL OPR W/ GEARBOX																																																																																																																												
					BALL					SW, CL800, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR										FLGD, RF, CL150, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB/TPE										507,517,518																																																																																																																		
										FLOATING BALL, LEVER OPR										TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX																																																																																																																												
					GLOBE					SW, CL800, API STD 602, A105N BODY, TRIM 16, OS&Y, BB, HNDWHL OPR										FLGD, RF, CL150, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 8, OS&Y, BB, SWIVEL PLUG DISC, HNDWHL OPR										507.516																																																																																																																		
VALVES (CONT.)					PLUG					SW, CL800, API STD 599, A105N BODY, WRENCH OPR										FLGD, RF, CL150, API STD 599, ASTM A216 WCB BODY										507																																																																																																																		
										WRENCH OPR										HNDWHL OPR W/ GEARBOX																																																																																																																												
					BUTTERFLY																																																																																																																																											
CHECK					SW, CL800, API STD 602, A105N BODY, TRIM 16, PISTON LIFT, BC										DBLE FLGD OR LUG, RF, CL150, API STD 594 TYPE A, ASTM A216 WCB BODY, DUAL PLATE										507,514,527,534																																																																																																																							
					TRIM 12										TRIM 8																																																																																																																																	
STRAINERS					Y-TYPE, SW, CL800, MAN STD DESIGN, A105N BODY, STLS STL SCRIN										I-TYPE, RF/BW, CL150/AS PER PIPE & FITTING SCH, ASME B16.5/B16.9, ASTM A105N, ASTM A234 WPB, STAINLESS STEEL SCREEN/PERFORATED SHEET																																																																																																																																	
GASKETS					FLAT RING, CL150, ASME B16.21, CNAF, 2 MM THICK										FLAT RING, CL150, ASME B16.21, CNAF, 3 MM THICK										300,301,305																																																																																																																							
					SPIRAL WOUND, LOW STRESS, CL150, ASME B16.20, SS 316 WINDING 4.5MM THICK, GRAPHITE FILLER, INNER AND CENTERING RING 3.2 MM THICK																				300.305																																																																																																																							
BOLTING					STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153																				350																																																																																																																							
BLANKS & SPACERS					FIG 8, RF, CL150, ASME B16.48, ASTM A516 GRADE 70N										SPADE & SPACER, RF, CL150, ASME B16.48, ASTM A516 GRADE 70N																																																																																																																																	
NOTES: 108 WALL THICKNESSES UP TO NPS 24 ARE ACCEPTABLE FOR FULL VACUUM @ 38°C 109 MILL TOLERANCE SHALL BE IN ACCORDANCE WITH ASME B 31.3 110 SHALL BE SUPPLIED WITH WELD COEFFICIENT OF 1 FOR SAW PROCESS 202 BRANCH CONNECTIONS SHALL BE IN ACCORDANCE WITH CORRESPONDING BRANCH TABLE IN APPENDIX 1 211 ULTRASONIC EXAMINATION (UT) METHOD SHALL NOT BE SUBSTITUTED FOR RADIOGRAPHIC EXAMINATION (RT) ON WELDED FITTINGS WITH WALL THICKNESS LESS THAN 12.7 MM 212 BENDING 5D RADIUS MAY BE USED FOR NPS 1.5 AND BELOW 250 SMOOTH FINISH FACING [3.2-6.3µm Ra] 300 SHALL BE IN ACCORDANCE WITH ASME B16.20 301 SELECTION OF FLAT GASKET SHALL BE BASED ON THE RANGE OF OPERATING CONDITIONS 305 FLAT RING AND SPIRAL WOUND GASKETS SHALL BE ASME CLASS 300 FOR ORIFICE FLANGE ASSEMBLY 350 BOLT DESIGN LOAD CALCULATION SHALL BE PERFORMED WHEN SPIRAL WOUND GASKETS ARE USED. THICKNESS OF HDG SHALL BE AS PER TABLE I OF ASTM 156 507 FIRE TESTED DESIGN 514 PISTON LIFT TYPE CHECK VALVES FOR HORIZONTAL INSTALLATION ONLY 516 VALVES 2" AND ABOVE SHALL BE SUPPLIED WITH REMOVABLE SEATS 517 ANTI STATIC DESIGN 518 SEAL MATERIALS SHALL BE IN ACCORDANCE WITH API 6D AND SELECTED USING OPERATING CONDITIONS 527 FOR CHECK VALVES NPS 2 AND ABOVE: API STD 594 TYPE B SWING TYPE WITH BOLTED COVER IS ACCEPTABLE IF DENSITY OF MEDIUM IS ABOVE 0.7 g/cm³ 534 FOR DUAL PLATE CHECK VALVE THE RETAINERLESS DESIGN IS MANDATORY FOR HYDROCARBON APPLICATIONS																																																																																																																																																

PIPING MATERIAL CLASS																												brass fertilizer							
FEED for Gas Gathering Pipelines & Associated Infrastructure(Phase 1)																																			
MATERIAL: CARBON STEEL														FLANGE RATING: ASME CLASS 150 RF								CLASS		B 06											
SERVICES: HYDROCARBON GAS HYDROCARBON LIQUID														PRESSURE / TEMPERATURE LIMITS (MATL GRP 1.1)										SHEET 1 OF 1				DATE: May-22		REVISION: 1					
														PRESS. (BARG)		19.6		19.6		17.7		13.8													
														TEMP. (°C)		-29		+38		+100		+200													
														CORROSION				6.0 mm				(103.207)													
NPS	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	24	26	28	30	32	34	36	38	40	42	44	46	NOTES							
SCHEDULE	XXS	XXS	XXS	XXS	XXS	160	120	XS	60	XS	40	40	40	30	XS	XS	XS	XS	60	30	30	30	-	-	-	-	80								
WALL THICK (mm)	7.47	7.82	9.09	10.15	5.49	6.02	7.11	12.70	12.70	13.50	14.80	16.90	18.10	20.10	24.20	26.20	28.20	30.20	32.20	34.20	36.20	38.20	40.30	42.30	15.88	15.88	17.48	109							
OD (mm)	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273	323.9	355.6	406.4	457	508	610	660	711	762	813	864	914	965	1016	1067	1118	1168								
PIPE	SEAMLESS, ASME B36.10, API SPEC 5L GRADE B PSL 2														WELDED, SAW 100% RADIOGRAPHY, ASME B36.10, API SPEC 5L GRADE B PSL 2														110						
ENDS	PE														BE																				
FLANGES	FLANGE	SW, RF, CL150, ASME B16.5, A105N														WN, RF, CL150, ASME B16.5, HUB THK AS PER PIPE SCHEDULE, ASTM A105N														250					
	BLIND	BLIND, RF, CL150, ASME B16.5, ASTM A105N														BLIND, RF, CL150, ASME B16.47 SERIES A, ASTM A105N														250					
	ORIFICE	ORIFICE, WN, RF, CL300, ASME B16.36, ASTM A105N																												250					
FITTINGS	ELBOW	45/90 DEG, SW, CL6000, ASME B16.11, A105N														45/90 DEG LR, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A234 WPB														211.212					
	TEE	TEE, SW, CL6000, ASME B16.11, A105N														EQUAL TEE/REDUCING TEE, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A234 WPB														202.211					
	REDUCER	RED CPLG, SW, CL6000, ASME B16.11, A105N														ECC/CONC, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A234 WPB														211					
	CAP	CAP, SW, CL6000, ASME B16.11, A105N														PIPE CAP, SEAMLESS, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A234 WPB																			
	OLET	SOCKOLET, SW, CL6000, ASME B16.11, MSS SP-97, A105N														WELDOLET, MSS SP-97, ASTM A105N														202					
	SWEEPOLET	SWEEPOLET, MANUFACTURER STANDARD DESIGN, ASTM A105N																												202					
VALVES	GATE	SW, CL800, API STD 602, A105N BODY, TRIM 16, OS&Y, BB, HNDWHL OPR														FLGD, RF, CL150, API STD 600, ASME B16.10, ASTM A216 WCB BODY, OS&Y, BB, FLEXIBLE WEDGE														507					
	BALL	SW, CL800, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR														FLGD, RF, CL150, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB/TPE														507,517,518					
																FLOATING BALL, LEVER OPR																			
	GLOBE	SW, CL800, API STD 602, A105N BODY, TRIM 16, OS&Y, BB, HNDWHL OPR														FLGD, RF, CL150, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 8, OS&Y, BB, SWIVEL PLUG DISC, HNDWHL OPR														507.516					
	PLUG	SW, CL800, API STD 599, A105N BODY, WRENCH OPR														FLGD, RF, CL150, API STD 599, ASTM A216 WCB BODY														507					
	BUTTERFLY	WRENCH OPR														HNDWHL OPR W/ GEARBOX																			
VALVES (CONT.)	CHECK	SW, CL800, API STD 602, A105N BODY, TRIM 16, PISTON LIFT, BC														DBLE FLGD OR LUG, RF, CL150, API STD 594 TYPE A, ASTM A216 WCB BODY, DUAL PLATE														507,514,527,534					
STRAINERS	Y-TYPE, SW, CL800, MAN STD DESIGN, A105N BODY, STLS STL SCRIN														T-TYPE, RF/BW, CL150/AS PER PIPE & FITTING SCH, ASME B16.5/B16.9, ASTM A105N, ASTM A234 WPB, STAINLESS STEEL SCREEN/PERFORATED SHEET																				
GASKETS	FLAT RING, CL150, ASME B16.21, CNAF, 2 MM THICK														FLAT RING, CL150, ASME B16.21, CNAF, 3 MM THICK														300,301,305						
	SPIRAL WOUND, LOW STRESS, CL150, ASME B16.20, SS 316 WINDING 4.5MM THICK, GRAPHITE FILLER, INNER AND CENTERING RING 3.2 MM THICK																												300.305						
BOLTING	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153																												350						
BLANKS & SPACERS	FIG 8, RF, CL150, ASME B16.48, ASTM A516 GRADE 70N														SPADE & SPACER, RF, CL150, ASME B16.48, ASTM A516 GRADE 70N																				
NOTES: 108 WALL THICKNESSES UP TO NPS 24 ARE ACCEPTABLE FOR FULL VACUUM @ 38°C 109 MILL TOLERANCE SHALL BE IN ACCORDANCE WITH ASME B 31.3 110 SHALL BE SUPPLIED WITH WELD COEFFICIENT OF 1 FOR SAW PROCESS 202 BRANCH CONNECTIONS SHALL BE IN ACCORDANCE WITH CORRESPONDING BRANCH TABLE IN APPENDIX 1 211 ULTRASONIC EXAMINATION (UT) METHOD SHALL NOT BE SUBSTITUTED FOR RADIOGRAPHIC EXAMINATION (RT) ON WELDED FITTINGS WITH WALL THICKNESS LESS THAN 12.7 MM 212 BENDING 5D RADIUS MAY BE USED FOR NPS 1.5 AND BELOW 250 SMOOTH FINISH FACING (3.2-6.3µm Ra) 300 SHALL BE IN ACCORDANCE WITH ASME B16.20 301 SELECTION OF FLAT GASKET SHALL BE BASED ON THE RANGE OF OPERATING CONDITIONS 305 FLAT RING AND SPIRAL WOUND GASKETS SHALL BE ASME CLASS 300 FOR ORIFICE FLANGE ASSEMBLY 350 BOLT DESIGN LOAD CALCULATION SHALL BE PERFORMED WHEN SPIRAL WOUND GASKETS ARE USED . THICKNESS OF HDG SHALL BE AS PER TABLE I OF ASTM 156 507 FIRE TESTED DESIGN 514 PISTON LIFT TYPE CHECK VALVES FOR HORIZONTAL INSTALLATION ONLY 516 VALVES 2" AND ABOVE SHALL BE SUPPLIED WITH REMOVABLE SEATS 517 ANTI STATIC DESIGN 518 SEAL MATERIALS SHALL BE IN ACCORDANCE WITH API 6D AND SELECTED USING OPERATING CONDITIONS 527 FOR CHECK VALVES NPS 2 AND ABOVE: API STD 594 TYPE B SWING TYPE WITH BOLTED COVER IS ACCEPTABLE IF DENSITY OF MEDIUM IS ABOVE 0.7 g/cm³ 534 FOR DUAL PLATE CHECK VALVE THE RETAINERLESS DESIGN IS MANDATORY FOR HYDROCARBON APPLICATIONS																																			



PIPING MATERIAL CLASS															FEED for Gas Gathering Pipelines & Associated Infrastructure(Phase 1)														
MATERIAL: CARBON STEEL, GALVANISED							FLANGE RATING:				ASME CLASS 150 RF				CLASS		B 13												
SERVICES: UTILITY WATER							PRESSURE / TEMPERATURE LIMITS (MATL GRP 1.1)								SHEET 1 OF 1		DATE: May-22		REVISION: 1										
							PRESS. (BARG)		19.6		19.6		17.7								13.8								
							TEMP. (°C)		-29		+38		+100								+200								
							CORROSION ALLOWANCE:				3.0 mm																		
NPS		0.5		0.75		1		1.5		2		3		4		6		8		10		12		14		16		NOTES	
SCHEDULE		XXS		XXS		XXS		XXS		160																38			
WALL THICK (mm)		7.47		7.82		9.09		10.15		5.49																			
OD (mm)		21.3		26.7		33.4		48.3		60.3																			
PIPE		SEAMLESS, ASME B36.10, API SPEC 5L GRADE B PSL 2, ASTM A153 CLASS B																											
ENDS		TE																											
FLANGES	FLANGE	THRD, RF, CL150, ASME B16.5, A105N, A153 CLASS B														250													
	BLIND	BLIND, RF, CL150, ASME B16.5, ASTM A105N, A153 CLASS B														250													
	ORIFICE															250													
FITTINGS	ELBOW	45/90 DEG, THRD, CL6000, ASME B16.11, A105N, A153 CLASS B																											
	TEE	TEE, THRD, CL6000, ASME B16.11, A105N, A153 CLASS B														202													
	REDUCER	RED CPLG, THRD, CL6000, ASME B16.11, A105N, A153 CLASS B																											
	CAP	CAP, THRD, CL6000, ASME B16.11, A105N, A153 CLASS B																											
	OLET	THREDOLET, THRD, CL6000, ASME B16.11, MSS SP-97, A105N, A153 CLASS B														202													
	SWEEPOLET																												
VALVES	GATE	THRD, CL800, API STD 602, A105N BODY, TRIM 16, OS&Y, BB, HNDWHL														507													
	BALL																												
	GLOBE	THRD, CL800, API STD 602, A105N BODY, TRIM 16, OS&Y, BB, SWIVEL														507													
	PLUG																												
	BUTTERFLY																												
	CHECK	THRD, CL800, API STD 602, A105N BODY, TRIM 16, PISTON LIFT, BC														507													
STRAINERS		Y-TYPE, THRD, CL800, MAN STD DESIGN, A105N BODY, A153 CLASS B, STL5 STL SCRNL																											
GASKETS		FLAT RING, CL150, ASME B16.21, CNAF, 2 MM THICK														300.301													
		SPIRAL WOUND, LOW STRESS, CL150, ASME B16.20, SS 316 WINDING 4.5MM THICK, GRAPHITE FILLER, INNER AND CENTERING RING 3.2 MM THICK														300													
BOLTING		STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153														350													
BLANKS & SPACERS		FIG 8, RF, CL150, ASME B16.48, ASTM A516 GRADE 70N																											
NOTES (CONTINUED OVERLEAF):																													
109 MILL TOLERANCE SHALL BE IN ACCORDANCE WITH ASME B31.3																													
202 BRANCH CONNECTIONS SHALL BE IN ACCORDANCE WITH CORRESPONDING BRANCH TABLE IN APPENDIX 1																													
250 SMOOTH FINISH FACING (3.2-6.3µm Ra)																													
300 SHALL BE IN ACCORDANCE WITH ASME B16.20																													
301 SELECTION OF FLAT GASKET SHALL BE BASED ON THE RANGE OF OPERATING CONDITIONS																													
350 BOLT DESIGN LOAD CALCULATION SHALL BE PERFORMED WHEN SPIRAL WOUND GASKETS ARE USED - HICKNESS OF HDG SHALL BE AS PER TABLE I OF ASTM 156																													
507 FIRE TESTED DESIGN																													

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PIPING MATERIAL CLASS																												brass fertilizer					
FEED for Gas Gathering Pipelines & Associated Infrastructure(Phase 1)																																	
MATERIAL: CARBON STEEL														FLANGE RATING: ASME CLASS 900 RJ										CLASS				C 03					
SERVICES:		HYDROCARBON GAS HYDROCARBON LIQUID												PRESSURE / TEMPERATURE LIMITS (MATL GRP 1.1)										SHEET 1 OF 1									
														PRESS. (BARG)														153.2		139.8		131.4	
														TEMP. (°C)														-29		+38		+100	
CORROSION														3.0 mm										DATE: May-22/REVISION: 1									
NPS		0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	24	26	28	30	32	34	36	38	40	42	44	46	NOTES				
SCHEDULE		160	160	160	160	160	160	120	120	120	120	XXS	120	120	100	100	100	-	-	-	-	-	-	-	-	-	-	-	-				
WALL THICK (mm)		4.78	5.56	6.35	7.14	5.49	6.02	7.11	12.70	12.70	13.50	14.80	16.90	18.10	20.10	24.20	26.20	28.20	30.20	32.20	34.20	36.20	38.20	40.30	42.30	53.10	55.40	57.80	109				
OD (mm)		21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273	323.9	355.6	406.4	457	508	610	660	711	762	813	864	914	965	1016	1067	1118	1168					
PIPE		SEAMLESS, ASME B36.10, API SPEC 5L GRADE B PSL 2														WELDED, SAW 100% RADIOGRAPHY, ASME B36.10, API SPEC 5L GRADE X60 PSL 2										110							
ENDS		PE												BE																			
FLANGES	FLANGE	SW, RJ, CL1500, ASME B16.5, A105N				WN, RJ, CL900, ASME B16.5, HUB THK AS PER PIPE SCHEDULE, ASTM A105N										WN, RJ, CL900, ASME B16.5, HUB THK AS PER PIPE SCHEDULE, ASTM A694 F60		WN, RJ, CL900, ASME B16.47 SERIES A, HUB THK AS PIPE SCHEDULE, ASTM A694 F60										252.261					
	BLIND	BLIND, RJ, CL900, ASME B16.5, ASTM A105N														BLIND, RJ, CL900, ASME B16.47 SERIES A, ASTM A105N										252.261							
	ORIFICE					ORIFICE, WN, RJ, CL900, ASME B16.36, ASTM A105N										ORIFICE, WN, RJ, CL900, ASME B16.36, ASTM A694												252					
FITTINGS	ELBOW	45/90 DEG, SW, CL6000, ASME B16.11, A105N				45/90 DEG LR, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A234 WPB										45/90 DEG LR, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A860 WPHY 60										212							
	TEE	TEE, SW, CL6000, ASME B16.11, A105N				EQUAL TEE/REDUCING TEE, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A234 WPB										EQUAL TEE/REDUCING TEE, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A860 WPHY 60										202							
	REDUCER	RED CPLG, SW, CL6000, ASME B16.11, A105N				ECC/CONC, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A234 WPB										ECC/CONC, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A860 WPHY 60																	
	CAP	CAP, SW, CL6000, ASME B16.11, A105N				PIPE CAP, SEAMLESS, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A234 WPB										PIPE CAP, SEAMLESS, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A860 WPHY 60																	
	OLET	SOCKET, SW, CL6000, ASME B16.11, MSS SP-97, A105N				WELDOLET, MSS SP-97, ASTM A105N																				202							
	SWEEPOLET					SWEEPOLET, MANUFACTURER STANDARD DESIGN, ASTM A105N										SWEEPOLET, MANUFACTURER STANDARD DESIGN, ASTM A694 F60										202							
VALVES	GATE	SW, CL1500, API STD 602, A105N BODY, TRIM 16, OS&Y, BB, HNDWHL OPR				FLGD, RJ, CL900, API STD 600, ASME B16.10, ASTM A216 WCB BODY, OS&Y, BB, FLEXIBLE WEDGE																				507							
						TRIM 16					TRIM 8																						
						HNDWHL OPR					HNDWHL OPR W/ GEARBOX																						
	BALL	SW, CL1500, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR				FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB/TPE, TRUNNION MTD BALL										507,514,518																	
VALVES (CONT.)	GLOBE	SW, CL1500, API STD 602, A105N BODY, TRIM 16, OS&Y, BB, HNDWHL OPR				FLGD, RJ, CL900, API STD 623, ASME															507.516												
						(529) HNDWHL OPR W/ GEARBOX																											
	PLUG	SW, CL1500, API STD 599, A105N BODY, WRENCH OPR				FLGD, RJ, CL900, API STD 599, ASTM A216 WCB BODY															507												
						WRENCH OPR HNDWHL OPR W/ GEARBOX																											
	BUTTERFLY																																
CHECK	SW, CL1500, API STD 602, A105N BODY, TRIM 16, PISTON LIFT, BC				DBLE FLGD OR LUG, RJ, CL900, API STD 594 TYPE A, ASTM A216 WCB BODY, DUAL PLATE										507,514,527,534																		
					TRIM 12					TRIM 8																							
STRAINERS		Y-TYPE, SW, CL1500, MAN STD DESIGN, A105N BODY, STL5 STL SCRIN				T-TYPE, RJ/BW, CL900/AS PER PIPE & FITTING SCH, ASME B16.5/B16.9, ASTM A105N, ASTM A234 WPB, STAINLESS STEEL SCREEN/PERFORATED SHEET																											
GASKETS		RING JOINT, OCTAGONAL, CL900, ASME B16.20, SOFT IRON																															
BOLTING		STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B7, A194 GRADE 2H, HOT DIP GALVANIZED (HDG) PER ASTM A153																															
BLANKS & SPACERS		FIG 8, RJ, CL900, ASME B16.48, ASTM A516 GRADE 70N								SPADE & SPACER, RJ, CL900, ASME B16.48, ASTM A516 GRADE 70N																							
NOTES: 109 MILL TOLERANCE SHALL BE IN ACCORDANCE WITHASME B 31.3 110 SHALL BE SUPPLIED WITH WELD COEFFICIENT OF 1 FOR SAW PROCESS 202 BRANCH CONNECTIONS SHALL BE IN ACCORDANCE WITH CORRESPONDING BRANCH TABLE IN APPENDIX 1 212 BENDING 5D RADIUS MAY BE USED FOR NPS 1.5 AND BELOW 252 HARDNESS OF RING JOINT GROOVE SHALL BE 20HB GREATER THAN GASKET HARDNESS 261 FOR FLANGE NPS 38 AND ABOVE, THE MANUFACTURER SHALL PROVIDE THE DIMENSIONS OF RING JOINT FACING (GROOVE AND RAISED PORTION) AND THE GASKET GEOMETRY 507 FIRE TESTED DESIGN 514 PISTON LIFT TYPE CHECK VALVES FOR HORIZONTAL INSTALLATION ONLY 516 VALVES 2" AND ABOVE SHALL BE SUPPLIED WITH REMOVABLE SEATS 517 ANTI STATIC DESIGN 518 SEAL MATERIALS SHALL BE IN ACCORDANCE WITH API 6D AND SELECTED USING OPERATING CONDITIONS 527 FOR CHECK VALVES NPS 2 AND ABOVE: API STD 594 TYPE B SWING TYPE WITH BOLTED COVER IS ACCEPTABLE IF DENSITY OF MEDIUM IS ABOVE 0.7 g/cm³ 529 HANDWHEEL OPERATED 534 FOR DUAL PLATE CHECK VALVE THE RETAINERLESS DESIGN IS MANDATORY FOR HYDROCARBON APPLICATIONS																																	

PIPING MATERIAL CLASS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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PIPING MATERIAL CLASS												brass fertilizer	
FEED for Gas Gathering Pipelines & Associated Infrastructure(Phase 1)													
MATERIAL: STAINLESS STEEL - UNS N08904 + TYPE 316L				FLANGE RATING: ASME CLASS 900 RJ				CLASS		C 50			
SERVICES:	CORROSION INHIBITOR DEMULSIFIER ANTI-FOAM			PRESSURE / TEMPERATURE LIMITS (MATL GRP 2.3)				SHEET 1	OF 1	REVISION: 1			
				PRESS. (BARG)		124.1 124.1 104.4 87.5							
				TEMP. (°C)		-46 +38 +100 +200							
				CORROSION 0 mm				DATE: May-22					
NPS	0.5	0.75	1	1.5	2				NOTES				
SCHEDULE	80S	80S	80S	80S	80S								
WALL THICK (mm)	3.73	3.91	4.55	5.08	7.11				24				
OD (mm)	21.3	26.7	33.4	48.3	60.3								
PIPE SEAMLESS, ASME B36.10/B36.19, ASTM A312 UNS N08904													
ENDS				PE				BE					
FLANGES	FLANGE	SW, RJ, CL1500, ASME B16.5, A182 F316L				WN, RJ, CL1500, ASME B16.5, HUB THK AS PER PIPE SCHEDULE, ASTM A182 F316L				252			
	BLIND ORIFICE	BLIND, RJ, CL1500, ASME B16.5, ASTM A182 F316L								251.252			
FITTINGS	ELBOW	45/90 DEG, SW, CL3000, ASME B16.11, A182 F316L											
	TEE	TEE, SW, CL3000, ASME B16.11, A182 F316L								202			
	REDUCER	RED CPLG, SW, CL3000, ASME B16.11, A182 F316L											
	CAP	CAP, SW, CL3000, ASME B16.11, A182 F316L											
	OLET	SOCKOLET, SW, CL3000, ASME B16.11, MSS SP-97, A182 F316L								202			
	SWEEPOLET												
VALVES	GATE												
	BALL	SW, CL1500, API SPEC 6D, A182 F316L BODY, SS316 TRIM, FLTING BALL, LEVER OPR				FLGD, RJ, CL1500, OR HUB CONNECTOR, API SPEC 6D, ASTM A182 F316/316L OR ASTM A351 CF8M BODY, SS316 TRIM, SPB/TPE, TRUNNION MTD BALL LEVER OPR				507,517,518			
	GLOBE	SW, CL1500, API STD 602, A182 F316L BODY, TRIM 12, OS&Y, BB, HNDWHL OPR				FLGD, RJ, CL1500, API STD 623, ASME B16.10, ASTM A351 CF8M BODY, TRIM 16, OS&Y, BB, SWIVEL PLUG DISC				507,516			
	NEEDLE	SW, CL1500, A182 F316L BODY, SS316 TRIM, BB, HNDWHL OPR								507			
	PLUG												
	CHECK	SW, CL1500, API STD 602, A182 F316L BODY, TRIM 16, PISTON LIFT, BC				DBLE FLGD OR LUG, RJ, CL1500, OR HUB CONNECTOR, API STD 594 TYPE A, ASTM A351 CF8M BODY, TRIM 12, DUAL PLATE				507,514			
STRAINERS		Y-TYPE, SW, CL1500, MAN STD DESIGN, A182 F316L BODY, STL5 STL SCRNI											
GASKETS		RING JOINT, OCTAGONAL, CL900, ASME B16.20, TYPE 316											
BOLTING		STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A193 GRADE B8M CL2, A194 GRADE 8MA											
BLANKS & SPACERS		FIG 8, RJ, CL900, ASME B16.48, ASTM A240 GRADE 316L											
NOTES:													
113 PIPE WALL THICKNESS CALCULATIONS ARE PERFORMED WITH ALLOWABLE STRESS VALUES OF ASTM A312 TP316L				514 PISTON LIFT TYPE CHECK VALVES FOR HORIZONTAL INSTALLATION ONLY 516 VALVES 2" AND ABOVE SHALL BE SUPPLIED WITH REMOVABLE SEATS									
202 BRANCH CONNECTIONS SHALL BE IN ACCORDANCE WITH CORRESPONDING BRANCH TABLE IN APPENDIX I				517 ANTI STATIC DESIGN									
251 ACCEPTABLE MATERIAL SUBSTITUTION: ASTM A182 F316				518 SEAL MATERIALS SHALL BE IN ACCORDANCE WITH API 6D AND SELECTED USING OPERATING CONDITIONS									
252 HARDNESS OF RING JOINT GROOVE SHALL BE 20HB GREATER THAN GASKET HARDNESS													
507 FIRE TESTED DESIGN													

PIPING MATERIAL CLASS																														brass fertilizer									
FEED for Gas Gathering Pipelines & Associated Infrastructure(Phase 1)																																							
MATERIAL: DUPLEX STAINLESS STEEL - UNS S31803															FLANGE RATING: ASME CLASS 900 RJ					CLASS					C 70														
SERVICES: HYDROCARBON GAS (H2S /CO2 HEAVY CORROSION) HYDROCARBON LIQUID (H2S /CO2 HEAVY CORROSION) CLOSED DRAIN															PRESSURE / TEMPERATURE LIMITS (MATL GRP 2.8)										SHEET 1 OF 1					REVISION: 1									
															PRESS. (BARG)					155.1		155.1		152											128				
															TEMP. (°C)					-46		+38		+100											+200				
															CORROSION 0 mm																								
NPS	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	24	26	28	30	32	34	36	38	40	42	44	46	NOTES											
SCHEDULE	80S	80S	80S	80S	80S	40S	40S	40S	80S	80S	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-											
WALL THICK (mm)	3.73	3.91	4.55	5.08	5.54	5.49	6.02	7.11	12.70	12.70	13.50	14.80	16.90	18.10	20.10	24.20	26.20	28.20	30.20	32.20	34.20	36.20	38.20	40.30	42.30	44.30	46.30	109											
OD (mm)	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273	323.9	355.6	406.4	457	508	610	660	711	762	813	864	914	965	1016	1067	1118	1168												
PIPE SEAMLESS/WELDED, 100% RADIOGRAPHY, ASME B36.10/B36.19, ASTM A790 S31803/ASTM A928 S31803 CL 1																														112									
ENDS PE BE																																							
FLANGES	FLANGE		SW, RJ, CL1500, ASME B16.5, A182 F51										WN, RJ, CL900, ASME B16.5, HUB THK AS PER PIPE SCHEDULE, ASTM A182 F51										WN, RJ, CL900, ASME B16.5, HUB THK AS PER PIPE SCHEDULE, ASTM A182 F51							252.261									
	BLIND		BLIND, RJ, CL900, ASME B16.5, ASTM A182 F51										BLIND, RJ, CL900, ASME B16.47 SERIES A, ASTM A182 F51																	252.261									
	ORIFICE		ORIFICE, WN, RJ, CL900, ASME B16.36, ASTM A182 F51																											252									
FITTINGS	ELBOW		45/90 DEG, SW, CL3000, ASME B16.11, A182 F51										45/90 DEG LR, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A815 S31803 WP-S/WP-WX																										
	TEE		TEE, SW, CL3000, ASME B16.11, A182 F51										EQUAL TEE/REDUCING TEE, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A815 S31803 WP-S/WP-WX																			202							
	REDUCER		RED CPLG, SW, CL3000, ASME B16.11, A182 F51										ECC/CONC, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A815 S31803 WP-S/WP-WX																										
	CAP		CAP, SW, CL3000, ASME B16.11, A182 F51										PIPE CAP, SEAMLESS, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A815 S31803 WP-S																										
	OLET		SOCKOLET, SW, CL3000, ASME B16.11, MSS SP-97, A182 F51										WELDOLET, MSS SP-97, ASTM A182 F51																			202							
	SWEEPOLET		SWEEPOLET, MANUFACTURER STANDARD DESIGN, ASTM A182 F51																													202							
VALVES	GATE		SW, CL1500, API STD 602, A182 F51 BODY, TRIM HF [Co-Cr ALLOY], OS&Y, BB, HNDWHL OPR										FLGD, RJ, CL900, API STD 600, ASME B16.10, ASTM A995 GR 4A BODY, UNS S31803 TRIM, OS&Y, BB, FLEXIBLE WEDGE																			507							
			TRIM HF [Co-Cr ALLOY]																																				
			HNDWHL OPR										HNDWHL OPR W/ GEARBOX																										
	BALL		SW, CL1500, API SPEC 6D, A182 F51 BODY, UNS S31803 TRIM, FLTING BALL										FLGD, RJ, CL900, API SPEC 6D, ASTM A182 F51 OR ASTM A995 GR 4A BODY, UNS S31803 TRIM, SPB/TPE, TRUNNION MTD BALL																			507,517,518,549							
			LEVER OPR										HNDWHL OPR W/ GEARBOX																										
	GLOBE		SW, CL1500, API STD 602, A182 F51 BODY, TRIM HF [Co-Cr ALLOY], OS&Y, BB, HNDWHL OPR										FLGD, RJ, CL900, API STD 623, ASME B16.10, ASTM A995 GR 4A BODY, TRIM HF [Co-Cr ALLOY], OS&Y, BB, SWIVEL PLUG DISC																				507.516						
			(529)										HNDWHL OPR W/ GEARBOX																										
	PLUG		SW, CL1500, API STD 599, A182 F51 BODY, WRENCH OPR										FLGD, RJ, CL900, API STD 599, ASTM A995 GR 4A BODY, UNS S31803 TRIM																				507						
WRENCH OPR										HNDWHL OPR W/ GEARBOX																													
BUTTERFLY																																							
CHECK		SW, CL1500, API STD 602, A182 F51 BODY, TRIM HF [Co-Cr ALLOY], PISTON LIFT, BC										DBLE FLGD OR LUG, RJ, CL900, API STD 594 TYPE A, ASTM A995 GR 4A BODY, TRIM HF [Co-Cr ALLOY], DUAL PLATE																			507,514,527,534								
STRAINERS			Y-TYPE, SW, CL1500, MAN STD DESIGN, A182 F51 BODY, UNS S31803 SCRIN										T-TYPE, RJ/BW, CL900/AS PER PIPE & FITTING SCH, ASME B16.5/B16.9, ASTM A815 S31803 WP-S, UNS S31803 SCREEN/PERFORATED SHEET																										
GASKETS			RING JOINT, OCTAGONAL, CL900, ASME B16.20, TYPE 316																																				
BOLTING			STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A453 GRADE 660 CLASS B																																				
BLANKS & SPACERS			FIG 8, RJ, CL900, ASME B16.48, ASTM A240 UNS S31803										SPADE & SPACER, RJ, CL900, ASME B16.48, ASTM A240 UNS S31803																										
NOTES: 3 IF PRESENCE OF H2S IN THE FLUID, MATERIAL SHALL COMPLY WITH NACE MR0175/ISO 15156-3 16 MATERIAL SHALL COMPLY WITH ASME B36.10/B36.19 109 MILL TOLERANCE SHALL BE IN ACCORDANCE WITH ASME B31.3 112 SHALL BE SUPPLIED WITH WELD COEFFICIENT OF 1 202 BRANCH CONNECTIONS SHALL BE IN ACCORDANCE WITH CORRESPONDING BRANCH TABLE IN APPENDIX 1 252 HARDNESS OF RING JOINT GROOVE SHALL BE 20HB GREATER THAN GASKET HARDNESS 261 FOR FLANGE NPS 38 AND ABOVE, THE MANUFACTURER SHALL PROVIDE THE DIMENSIONS OF RING JOINT FACING (GROOVE AND RAISED PORTION) AND THE GASKET GEOMETRY 507 FIRE TESTED DESIGN 514 PISTON LIFT TYPE CHECK VALVES FOR HORIZONTAL INSTALLATION ONLY 516 VALVES 2" AND ABOVE SHALL BE SUPPLIED WITH REMOVABLE SEATS 517 ANTI STATIC DESIGN 518 SEAL MATERIALS SHALL BE IN ACCORDANCE WITH API 6D AND SELECTED USING OPERATING CONDITIONS 527 FOR CHECK VALVES NPS 2 AND ABOVE: API STD 594 TYPE B SWING TYPE WITH BOLTED COVER IS ACCEPTABLE IF DENSITY OF MEDIUM IS ABOVE 0.7 g/cm³ 529 HANDWHEEL OPERATED 534 FOR DUAL PLATE CHECK VALVE THE RETAINERLESS DESIGN IS MANDATORY FOR HYDROCARBON APPLICATIONS 549 MATERIAL SUBSTITUTION CARBON STEEL WITH CRA WELD OVERLAY ON ALL WETTED SURFACES MAY BE CONSIDERED UNDER SPECIFIC CONDITIONS																																							

PIPING MATERIAL CLASS																							brass fertilizer	
FEED for Gas Gathering Pipelines & Associated Infrastructure(Phase 1)																								
MATERIAL: DUPLX STAINLESS STEEL - UNS S31803											FLANGE RATING: ASME CLASS 1500 RJ.				CLASS		D 70							
SERVICES: HYDROCARBON GAS (H2S / CO2 HEAVY CORROSION) HYDROCARBON LIQUID (H2S / CO2 HEAVY CORROSION) CLOSED DRAIN											PRESSURE / TEMPERATURE LIMITS (MATL GRP 2.8)													
											PRESS. (BARG)		258.6		258.6		253.3		213.3		SHEET 1 OF 1			
											TEMP. (°C)		-46		+38		+100		+200		REVISION: 1			
											CORROSION 0 mm				DATE: May-22									
NPS	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	24	26	28	30	32	34	36	NOTES	
SCHEDULE	80S	80S	80S	80S	80S	80S	80S	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
WALL THICK (mm)	3.73	3.91	4.55	5.08	5.54	7.62	8.56	11.50	14.90	18.60	22.10	24.20	27.70	29.60	32.90	39.50	42.80	46.10	49.40	52.6	55.9	59.2	109	
OD (mm)	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273	323.9	355.6	406.4	457	508	610	660	711	762	813	864	914		
PIPE	SEAMLESS/WELDED, 100% RADIOGRAPHY, ASME B36.10/B36.19, ASTM A790 S31803/ASTM A928 S31803 CL 1																						112	
ENDS	PE		BE																					
FLANGES & HUB	FLANGE	SW, RJ, CL1500, ASME B16.5, A182 F51						WN, RJ, CL1500, ASME B16.5, HUB THK AS PER PIPE SCHEDULE, ASTM A182 F51						WN, RJ, CL900, ASME B16.5, HUB THK AS PER PIPE SCHEDULE, ASTM A182 F51						252				
	BLIND	BLIND, RJ, CL1500, ASME B16.5, ASTM A182 F51										BLIND, RJ, CL900, ASME B16.47 SERIES A, ASTM A182 F51												252
	ORIFICE	ORIFICE, WN, RJ, CL1500, ASME B16.36, ASTM A182 F51																						252
FITTINGS	ELBOW	45/90 DEG, SW, CL6000, ASME B16.11, A182 F51						45/90 DEG LR, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A815 S31803 WP-S/WP-WX																
	TEE	TEE, SW, CL6000, ASME B16.11, A182 F51						EQUAL TEE/REDUCING TEE, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A815 S31803 WP-S/WP-WX																202
	REDUCER	RED CPLG, SW, CL6000, ASME B16.11, A182 F51						ECC/CONC, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A815 S31803 WP-S/WP-WX																
	CAP	CAP, SW, CL6000, ASME B16.11, A182 F51						PIPE CAP, SEAMLESS, ASME B16.9, THK AS PER PIPE SCHEDULE, ASTM A815 S31803 WP-S																
	OLET	SOCKOLET, SW, CL6000, ASME B16.11, MSS SP-97, A182 F51						WELDOLET, MSS SP-97, ASTM A182 F51																202
	SWEEPOLET							SWEEPOLET, MANUFACTURER STANDARD DESIGN, ASTM A182 F51																202
VALVES	GATE	SW, CL1500, API STD 602, A182 F51 BODY, TRIM HF [Co-Cr ALLOY], OS&Y, BB, HNDWHL OPR						FLGD, RJ, CL1500, API STD 600, ASME B16.10, ASTM A995 GR 4A BODY, UNS S31803 TRIM, OS&Y, BB, FLEXIBLE WEDGE																507
		TRIM HF [Co-Cr ALLOY]																						
		HNDWHL OPR						HNDWHL OPR W/ GEARBOX																
	BALL	SW, CL1500, API SPEC 6D, A182 F51 BODY, UNS S31803 TRIM, FLTING BALL, LEVER OPR						FLGD, RJ, CL1500, API SPEC 6D, ASTM A182 F51 OR ASTM A995 GR 4A BODY, UNS S31803 TRIM, SPB/TPE, TRUNNION MTD BALL																507,517,518,549
LEVER OPR						HNDWHL OPR W/ GEARBOX																		
VALVES (CONT.)	GLOBE	SW, CL1500, API STD 602, A182 F51 BODY, TRIM HF [Co-Cr ALLOY], OS&Y, BB, HNDWHL OPR						FLGD, RJ, CL1500, API STD 623, ASME B16.10, ASTM A995 GR 4A BODY, TRIM HF [Co-Cr ALLOY], OS&Y, BB, SWIVEL PLUG DISC												507.516				
								(529) HNDWHL OPR W/ GEARBOX																
	PLUG	SW, CL1500, API STD 599, A182 F51 BODY, WRENCH OPR						FLGD, RJ, CL1500, API STD 599, ASTM A995 GR 4A BODY, UNS S31803 TRIM												507.000				
								(531) HNDWHL OPR W/ GEARBOX																
	BUTTERFLY																							
CHECK	SW, CL1500, API STD 602, A182 F51 BODY, TRIM HF [Co-Cr ALLOY], PISTON						DBLE FLGD OR LUG, RJ, CL1500, API STD 594 TYPE A, ASTM A995 GR 4A BODY, TRIM HF [Co-Cr ALLOY], DUAL PLATE																507,514,527,534,	
STRAINERS	Y-TYPE, SW, CL1500, MAN STD DESIGN, A182 F51 BODY, UNS S31803 SCRIN								T-TYPE, RJ/BW, CL1500/AS PER PIPE & FITTING SCH, ASME B16.5/B16.9, ASTM A182 F51, ASTM A815 S31803 WP-S/WP-WX, UNS S31803 SCREEN/PERFORATED SHEET															
GASKETS	RING JOINT, OCTAGONAL, CL1500, ASME B16.20, TYPE 316																							
BOLTING	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A453 GRADE 660 CLASS B																							
BLANKS & SPACERS	FIG 8, RJ, CL1500, ASME B16.48, ASTM A240 UNS S31803								SPADE & SPACER, RJ, CL1500, ASME B16.48, ASTM A240 UNS S31803															
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202 BRANCH CONNECTIONS SHALL BE IN ACCORDANCE WITH CORRESPONDING BRANCH TABLE IN APPENDIX 1											531 WRENCH OPERATED													
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507 FIRE TESTED DESIGN																								
514 PISTON LIFT TYPE CHECK VALVES FOR HORIZONTAL INSTALLATION ONLY																								
516 VALVES 2" AND ABOVE SHALL BE SUPPLIED WITH REMOVABLE SEATS											549 MATERIAL SUBSTITUTION CARBON STEEL WITH CRA WELD OVERLAY ON ALL WETTED SURFACES MAY BE CONSIDERED UNDER SPECIFIC CONDITIONS													
517 ANTI STATIC DESIGN																								

APPENDIX 1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
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